

Economic Shocks and Civil Conflict in Africa: A Re-examination of the Miguel-Satyanath-Sergenti Rainfall-Instrument Approach

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KU Leuven: D0S91a Advanced Applied Econometrics

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May 2026

Abstract

We revisit the seminal instrumental-variables (IV) study of Miguel et al. (2004), which uses annual rainfall growth as an exogenous shifter of per-capita GDP growth to identify the causal effect of economic conditions on civil conflict in 41 sub-Saharan African countries during 1981 until 1999. Pooled OLS recovers the qualitative direction of the original association that negative growth coincides with more conflict, $\hat{\beta} = -0.51$ ($p = 0.02$), but a battery of modern diagnostics flags the rainfall instruments as weak in our preferred two-way fixed-effects setting (cluster-robust first-stage $F = 2.89$, well below the Stock-Yogo 10%-maximum-size threshold of 19.93). Weak-instrument robust inference (Anderson and Rubin, 1949) delivers a 95% confidence interval of $[-4.41, 7.49]$, uninformative about the sign of the structural parameter. Restoring the country-specific linear time trends used by MSS in place of year fixed effects almost doubles the first-stage F to 4.45 and produces a 2SLS point estimate of -1.30 , in line with the OLS sign though still imprecisely estimated. Most decisively, a dynamic-panel Arellano-Bond specification (Arellano and Bond, 1991), which explicitly accounts for the strong persistence of conflict, yields a negative and statistically significant coefficient on growth (-0.29 , $p = 0.046$), corroborating the qualitative MSS conclusion while making clear that the rainfall instrument alone is fragile. The Wu-Hausman test also fails to reject exogeneity ($p = 0.71$), suggesting that the well-determined OLS estimate may be the most defensible point estimate in this sample.

1 Introduction

Whether adverse economic shocks trigger civil conflict is a question with first-order policy implications for the design of aid, climate adaptation programmes and post-conflict reconstruction. Ordinary regressions of conflict on contemporaneous growth confound at least three sources of endogeneity: reverse causation (conflict destroys output), measurement error in low-income GDP statistics, and omitted country traits correlated with both. Miguel et al. (2004) address all three by instrumenting GDP-per-capita growth with year-on-year rainfall growth, arguing that in agriculture-dependent African economies rainfall fluctuations are quasi-random shifters of output that satisfy both relevance and the exclusion restriction.

The MSS finding that a five-percentage-point decline in growth raises the probability of civil conflict by some twelve percentage points has been hugely influential, but has also drawn scrutiny. Ciccone (2011) argues that the link works through rainfall *levels* rather than growth, and Sarsons (2015) documents that rainfall predicts conflict downstream of dams that physically break the agricultural-income channel, casting doubt on the exclusion restriction.

This paper has two purposes. First, we conduct a careful applied econometric replication of Miguel et al. (2004) that systematically applies the obtained diagnostics: heteroscedasticity tests, fixed- versus random-effects choice, weak-instrument tests, overidentification tests, weak-IV-robust inference, cluster bootstraps, and a control-function probit. Second, recognising that conflict is highly persistent and that the rainfall instruments are fragile, we extend the analysis to a dynamic-panel system-GMM estimator (Arellano and Bond, 1991; Blundell and Bond, 1998) that exploits the panel’s longitudinal structure rather than relying on external instruments. Our central message is methodological: the MSS conclusion receives qualified support, but the rainfall instrument is too fragile to provide compelling causal evidence on its own. The contrast between specifications is instructive for any applied researcher confronting a single, marginally relevant instrument.

2 Conceptual framework

We adopt the simple “opportunity-cost” mechanism of Grossman (1991) and Chassang and Padró i Miquel (2009). Individuals decide between productive labour, with expected wage w proportional to current income y , and predation, with expected return π that is largely independent of y . Whenever $w(y) < \pi$ predation strictly dominates; the conflict-participation margin is therefore decreasing in y . Taking logarithms and first differences yields the linear approximation

$$C_{it} = \alpha_i + \delta_t + \beta g_{it}^y + \gamma X_{it} + u_{it}, \quad (1)$$

where $C_{it} = \mathbf{1}\{\text{conflict}_{it}\}$ is a binary indicator of civil conflict in country i and year t , $g_{it}^y = \Delta \ln(y_{it})$ is annual per-capita GDP growth, X_{it} collects time-varying controls (here lagged democracy/Polity 2), and α_i , δ_t are country and year fixed effects. Theory predicts $\beta < 0$.

The empirical challenge is identification of β : g_{it}^y is plausibly endogenous to u_{it} . Miguel et al. (2004)’s solution is to instrument g_{it}^y with rainfall growth g_{it}^R and its one-period lag. Identification requires (i) *relevance*: rainfall shifts agricultural output and hence GDP growth, and (ii) *validity/exclusion?*: rainfall affects conflict only through GDP. Both assumptions are testable to varying degrees; we report a range of formal diagnostics below.

3 Data

We use the replication panel distributed by Edward Miguel through the Harvard Dataverse ([doi:10.7910/DVN/27324](https://doi.org/10.7910/DVN/27324)), covering 41 sub-Saharan African countries from 1981 to 1999. After constructing one lag of GDP growth and dropping country-years with missing controls, our estimation sample is 702 country-year observations across 41 countries and 18 years (an unbalanced panel). The dependent variable `any_prio` is a PRIO indicator equal to one if a country experiences at least 25 battle deaths in a given year. The endogenous regressor is annual GDP-per-capita

growth `gdp_g`; the excluded instruments are current and lagged rainfall growth `GPCP_g` and `GPCP_g_1`, constructed from the Global Precipitation Climatology Project. Controls follow MSS: initial log income, lagged Polity 2, ethnic and religious fractionalisation, an oil-exporter dummy, log mountainous terrain, and log lagged population. Table 1 reports descriptive statistics. The unconditional conflict frequency is 26.9%, average growth is slightly negative (-0.6% p.a.), and the panel is democracy-poor (mean Polity 2 of -3.5).

Table 1: Descriptive statistics, estimation sample ($N = 702$)

Variable	N	Mean	SD	Min	Max
<code>any_prio</code>	702	0.269	0.444	0.000	1.000
<code>gdp_g</code>	702	-0.006	0.066	-0.474	0.319
<code>GPCP_g</code>	702	0.015	0.210	-0.550	1.680
<code>y_0</code>	702	1.16	0.90	0.316	4.84
<code>polity2l</code>	702	-3.55	5.55	-10	9
<code>ethfrac</code>	702	0.655	0.237	0.036	0.925
<code>relfrac</code>	702	0.487	0.185	0.000	0.783
<code>Oil</code>	702	0.117	0.321	0	1
<code>lmtnest</code>	702	1.58	1.43	0	4.42
<code>lpopl1</code>	702	8.77	1.20	5.75	11.7

4 Econometric specification and methods

Our reasoning proceeds in five steps that mirror the choices recommended in Wooldridge (2010).

(1) Pooled OLS as benchmark. We first estimate equation (1) by Pooled OLS with the full set of (time-invariant) controls. Because the outcome is binary this is a Linear Probability Model (LPM) and is heteroscedastic by construction with $\text{Var}(u_{it}) = p_{it}(1 - p_{it})$. We test heteroscedasticity formally with the Breusch–Pagan test and a White test in the Wooldridge (2010) simplified form, and always report cluster-robust standard errors clustered at the country level.

(2) Fixed versus random effects. The within transformation absorbs the time-invariant controls (`y_0`, `ethfrac`, `relfrac`, `Oil`, `lmtnest`, `lpopl1`); we therefore restrict the FE and RE specifications to time-varying covariates and add year fixed effects. The presence of country effects is checked with an F -test, and we report the Hausman test of the orthogonality of the RE-assumed effects to the regressors.

(3) Two-stage least squares. We then implement the MSS IV approach with $\{\text{GPCP}_g, \text{GPCP}_{g_1}\}$ as excluded instruments for `gdp_g`. We report the cluster-robust first-stage F on the excluded instruments and compare it to the Stock-Yogo critical value (19.93 for 10% maximum size with two IVs and one endogenous regressor). We additionally report the Wu-Hausman endogeneity

test, the Sargan/Hansen overidentification test, and the weak-IV-robust Anderson–Rubin (AR) confidence set, which retains correct coverage under arbitrarily weak first stages.

(4) Robustness: country trends. Because our two-way FE specification produces a markedly weaker first stage than the original MSS paper, we re-estimate the 2SLS adding country-specific linear time trends in place of year fixed effects, the original MSS choice. This sharpens identification by removing slow country-level secular movements (democratisation, technology diffusion) without absorbing the rainfall variation.

(5) Dynamic robustness: control function and System-GMM. Two further specifications relax the static-binary form. First, we re-estimate (1) as a probit with a Rivers-Vuong control function, plugging in the first-stage residuals as an extra regressor and reading the z -statistic on the residual as a Hausman-style test of exogeneity (Wooldridge, 2010). We prefer this approach to direct IV-Probit maximum likelihood because, with 41 country dummies and $T = 18$, the latter is exposed to incidental-parameters bias. Second, recognising that conflict is highly serially correlated, we estimate dynamic specifications including $C_{i,t-1}$ as a regressor. Within-FE on a dynamic panel suffers from Nickell (1981) bias of order $O(1/T)$, which with $T = 18$ remains non-trivial; we therefore use difference-GMM (Arellano and Bond, 1991) and system-GMM (Blundell and Bond, 1998) estimators with collapsed instrument matrices and lag depth two through three to keep the instrument count well below $N = 41$ (Roodman, 2009). Two-step robust standard errors are reported. We assess validity with the Sargan test on overidentifying restrictions and the Arellano-Bond AR(1) and AR(2) tests for autocorrelation in the differenced residuals.

5 Results

5.1 Main estimates

Table 2 collects the seven static specifications. Pooled-OLS column (1) reproduces the headline MSS direction: a one standard-deviation drop in growth (≈ 6.6 percentage points) raises the conflict probability by roughly $0.51 \times 0.066 \approx 3.4$ percentage points, statistically significant at the 5% level. Two-way fixed-effects column (2) and random-effects column (3) shrink the coefficient towards zero ($\hat{\beta} = -0.46, -0.44$) with slightly widened standard errors, as expected when we strip out between-country variation. The Hausman test fails to reject RE ($p = 0.998$); but the F -test for country effects is large ($F = 14.9, p < 0.001$); given the strong rejection of pooling, we maintain FE as the preferred estimator. The reduced-form column (4) shows that rainfall growth enters the conflict equation only weakly (lagged rainfall has $p = 0.08$); the first stage column (5) confirms that current-year rainfall growth has a statistically significant but quantitatively modest effect on GDP growth.

Column (6), our baseline 2SLS, is the most striking result. The point estimate on `gdp_g` is $+0.14$ with a clustered standard error of 1.33: the wrong sign, no statistical significance, and a confidence interval covering essentially all economically relevant magnitudes. This collapse of the IV estimate relative to the well-determined POLS estimate is the canonical fingerprint of a

weak instrument. Column (7) re-estimates the same model with country FE plus country-specific linear time trends—the MSS specification—and yields $\hat{\beta} = -1.30$ (SE 1.07, $p = 0.23$): the correct (negative) sign, larger in magnitude than OLS, but still imprecisely estimated. The qualitative direction of the original MSS finding is therefore highly sensitive to whether secular country-level trends are modelled.

	POLS (LPM)	FE 2-way	RE 2-way	Reduced form	First stage	2SLS	2SLS (trends)
GDP growth	−0.507** (0.222)	−0.460** (0.220)	−0.443** (0.217)			0.141 (1.330)	−1.297 (1.071)
GDP growth (lag)	0.020 (0.243)	−0.173 (0.221)	−0.121 (0.277)			−0.165 (0.229)	0.040 (0.185)
Rainfall growth				0.024 (0.055)	0.042** (0.018)		
Rainfall growth (lag)				−0.084* (0.048)	0.025 (0.016)		
Polity2 (lag)	0.000 (0.004)	−0.004 (0.005)	−0.002 (0.005)	−0.003 (0.005)	−0.001* (0.001)	−0.003 (0.006)	−0.004 (0.005)
Num.Obs.	702	702	702	702	702	702	702
R2	0.129	0.012	0.006	0.005	0.019	0.569	0.710
R2 Within						0.003	0.371
FE: ccode						X	X
FE: year						X	
Controls (POLS only)	Yes	—	—	—	—	—	—

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Cluster-robust SE at country level in parentheses. Models 1, 5 and 6 use 2-way FE; model 7 uses country FE plus country-specific linear time trends. Time-invariant controls included in POLS only.

Table 2: Effect of GDP growth on civil conflict in Africa, 1981–1999. Models (1)–(6) include year fixed effects (where applicable); model (7) replaces year FE with country-specific linear time trends. Cluster-robust SE in parentheses.

5.2 Specification diagnostics

Table 3 reports the formal diagnostics, grouped by what they test.

Heteroscedasticity. The Breusch–Pagan test overwhelmingly rejects ($\chi^2 = 142.5$, $p < 10^{-16}$), confirming the necessity of robust standard errors in the LPM.

Panel structure. The F -test of country fixed effects strongly rejects the null that they are

jointly zero ($F = 14.9$, $p < 10^{-16}$); the Hausman test does not distinguish FE from RE ($p = 0.998$). The classical Hausman statistic is, however, unreliable in this setting: its construction assumes that the FE estimator is efficient under H_0 , which fails once standard errors are clustered, and the test also has limited power in moderately unbalanced panels. We therefore default to FE following Wooldridge (2010).

Weak instruments. The cluster-robust first-stage F on the two excluded instruments is just 2.89 in our baseline 2-way FE specification, far below the Staiger and Stock (1997) rule-of-thumb of 10 or the Stock–Yogo (Stock and Yogo, 2005) 10%-maximum-size threshold of 19.93 (the corresponding 10%-relative-bias critical value is ≈ 11 , which the baseline F also fails to meet). Switching to the MSS-style country-trend specification lifts the F to 4.45 (cluster-robust Wald) or 7.48 (homoscedastic F), notably stronger but still below conventional thresholds.

Endogeneity. The Wu-Hausman exogeneity test fails to reject ($\chi^2 = 0.14$, $p = 0.71$); the control-function probit residual $\hat{v}$ is statistically insignificant ($z = -0.73$, $p = 0.46$). In this sample we have no positive evidence that OLS is inconsistent.

Overidentification. The Sargan test for the two-instrument design does not reject ($\chi^2 = 2.40$, $p = 0.12$), but with only one overidentifying restriction this test has low power and provides only weak reassurance about the exclusion restriction.

Weak-IV-robust inference. The Anderson-Rubin 95% confidence set for β is $[-4.41, 7.49]$, essentially uninformative about either the sign or the magnitude of the structural parameter. A non-parametric cluster bootstrap (500 replications, resampling countries with replacement) yields a closely comparable percentile interval of $[-3.52, 3.51]$.

Table 3: Specification and identification diagnostics

Statistic	Value	<i>p</i> -value
<i>Heteroscedasticity</i>		
Breusch–Pagan	142.52	$< 10^{-16}$
<i>Panel structure</i>		
<i>F</i> -test country FE ($H_0: c_i = 0$)	14.90	$< 10^{-16}$
Hausman FE vs RE	0.04	0.998
<i>Weak instruments</i>		
First-stage <i>F</i> baseline (cluster-robust)	2.89	0.056
First-stage <i>F</i> MSS-trends (cluster-robust Wald)	4.45	0.012
First-stage <i>F</i> MSS-trends (homoscedastic)	7.48	0.0006
Stock–Yogo 10%-maximum-size critical value	19.93	—
<i>Endogeneity</i>		
Wu–Hausman (AER <code>ivreg</code>)	0.14	0.705
Control-function probit $z(\hat{v})$	−0.73	0.464
<i>Overidentification</i>		
Sargan (AER <code>ivreg</code>)	2.40	0.121
<i>Weak-IV-robust inference</i>		
Anderson–Rubin 95% CI for β	[−4.41, 7.49]	—
Cluster-bootstrap 95% CI for β^{IV}	[−3.52, 3.51]	—

5.3 Robustness: dynamic panel-data estimators

The dynamic-panel results in Table 4 are the most encouraging for the substantive MSS conclusion. The lagged conflict indicator carries a large and highly significant coefficient (0.63, $p < 0.001$ in system-GMM), confirming substantial conflict persistence and indicating that the static specifications are mis-specified. Conditional on this dynamic structure, the Arellano-Bond estimator delivers $\hat{\beta} = -0.29$ with $p = 0.046$; the Blundell-Bond system-GMM yields a similar point estimate $\hat{\beta} = -0.30$ but no longer statistically significant ($p = 0.19$). The Sargan tests do not reject ($p = 0.51$ and 0.09 respectively); AR(2) tests do not detect second-order autocorrelation in differences ($p = 0.50$ and 0.47); both validations support the identifying restrictions. The qualitative MSS result: a negative relationship between growth and conflict, thus reappears once the dynamic structure of conflict is properly modelled, even without an external instrument.

	Arellano-Bond	Blundell-Bond
Conflict (lag)	0.334** (0.142)	0.628*** (0.095)
GDP growth	−0.293* (0.150)	−0.296* (0.155)
GDP growth (lag)	0.044 (0.211)	−0.046 (0.225)
Polity2 (lag)	0.000 (0.004)	−0.006* (0.003)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Two-step robust SE. Collapsed instrument matrix with lags 2–3.

Table 4: Dynamic panel-data estimators of conflict on growth. Two-step robust standard errors. Collapsed instrument matrix with lag depth 2–3.

6 Discussion and conclusion

This essay illustrates a tension that recurs in applied microeconometrics: a story that is compelling, theoretically well motivated and empirically detectable in OLS does not always survive strict instrumental-variables identification. Four lessons stand out.

First, the rainfall instruments are markedly weaker in our two-way FE specification ($F = 2.89$) than under the MSS country-trends design ($F = 4.45$). This is a useful reminder that the strength of an instrument is specification-dependent, and that seemingly small modelling choices (year fixed effects versus country trends) materially change identification.

Second, when an instrument is weak, conventional 2SLS t -statistics are unreliable and the only valid inference is weak-IV-robust (Anderson and Rubin, 1949; Staiger and Stock, 1997). Reporting only the 2SLS point estimate would have been misleading: our baseline 2SLS estimate is positive while every other reasonable specification points negative. The Anderson-Rubin confidence set, which makes no strength-of-identification assumption, is consequently very wide.

Third, the Wu-Hausman test and the control-function probit residual both fail to reject exogeneity of GDP growth. Combined with the weak first stage, this would justify reporting the well-determined POLS estimate of -0.51 as the most defensible point estimate in this sample, treating the IV strategy as an inconclusive robustness check rather than the primary identification.

Fourth, leveraging the panel dimension via dynamic-panel GMM offers an identification route that does not depend on an external instrument at all. Conditional on lagged conflict, both Arellano-Bond and Blundell-Bond recover negative growth coefficients near -0.30 , the first significant at the 5% level. Our overall reading is that MSS’s substantive conclusion, namely that adverse economic shocks raise the probability of civil conflict in Africa, receives qualified support, but the rainfall instrument is too fragile to provide compelling causal evidence on its own.

Limitations of our analysis include the binary measure of conflict (LPM versus probit/logit), the focus on contemporaneous shocks (richer distributed lags are plausible), and the omission of conflict-spillover spatial dynamics emphasised in subsequent literature. Each of these would be natural extensions for a longer-form study.

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